



# Dr. N.G.P. INSTITUTE OF TECHNOLOGY

(An Autonomous Institution)

Approved by AICTE, New Delhi & Affiliated to Anna University, Chennai)  
Recognized by UGC & Accredited by NAAC with A+ and NBA (BME, CSE, ECE, EEE and Mech)

## Department of Electrical and Electronics Engineering

### 22UEE407 – MINI PROJECT 1

#### FINAL REVIEW

***TITLE :IoT-ENABLED AUTONOMOUS LAWN MOWER WITH SELF-RECHARGING***

#### ***Supervisor***

Dr.SHRIRAM SRINIVASARANGAN RANGARAJAN,  
Head of the Department.

#### ***Team Members***

|            |              |
|------------|--------------|
| KISHOR S   | 710723105027 |
| MATHAVAN S | 710723105031 |
| MITHUN R   | 710723105032 |
| VIGNESH S  | 710723105060 |

# OBJECTIVES

- Design a system capable of autonomously cutting grass within a defined boundary without human intervention.
- Implement sensors to detect obstacles (like trees, rocks, or pets) and avoid collisions during operation.
- Optimize power usage, with options for battery management or solar assistance for extended operation.
- Provide a simple user interface (mobile app) for starting, stopping, and monitoring the mower status.
- Enable the mower to return to a charging dock when battery levels are low, supporting continuous autonomous operation.

# PROBLEM STATEMENT

- Traditional lawn mowers are manually operated, consume fossil fuels, and require significant human effort.
- They contribute to environmental pollution, increase maintenance costs, and may not be accessible for elderly or disabled users.
- A smart, rechargeable, and IoT-integrated solution is needed to enhance efficiency, reduce emissions, and allow remote operation.

# LITERATURE SURVEY

| S.NO | AUTHOR                                          | TITLE OF THE PAPER                                               | YEAR OF PUBLISH | DESCRIPTION                                                                                                                                                                   |
|------|-------------------------------------------------|------------------------------------------------------------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.   | M. Johnson, L. Smith, R. Patel, and K. Thompson | Design and Development of an Autonomous Lawn Mower Robot         | 2020            | We observed that automated mowing reduces human effort. This system uses ultrasonic sensors for obstacle detection and GPS for area coverage, improving lawn care efficiency. |
| 2.   | A. Gupta, N. Sharma, P. Kumar, and S. Reddy     | IoT-Based Smart Lawn Mower for Automated Grass Cutting           | 2021            | We observed that integrating IoT improves remote control and monitoring. This mower uses Wi-Fi modules and mobile app control for flexible operation and user convenience.    |
| 3.   | J. Brown, T. Wilson, H. Clark, and D. Martinez  | Solar-Powered Autonomous Lawn Mower with Efficient Path Planning | 2023            | We observed that solar power enhances energy efficiency. The system includes solar charging and AI-based path planning to ensure complete and eco-friendly mowing.            |

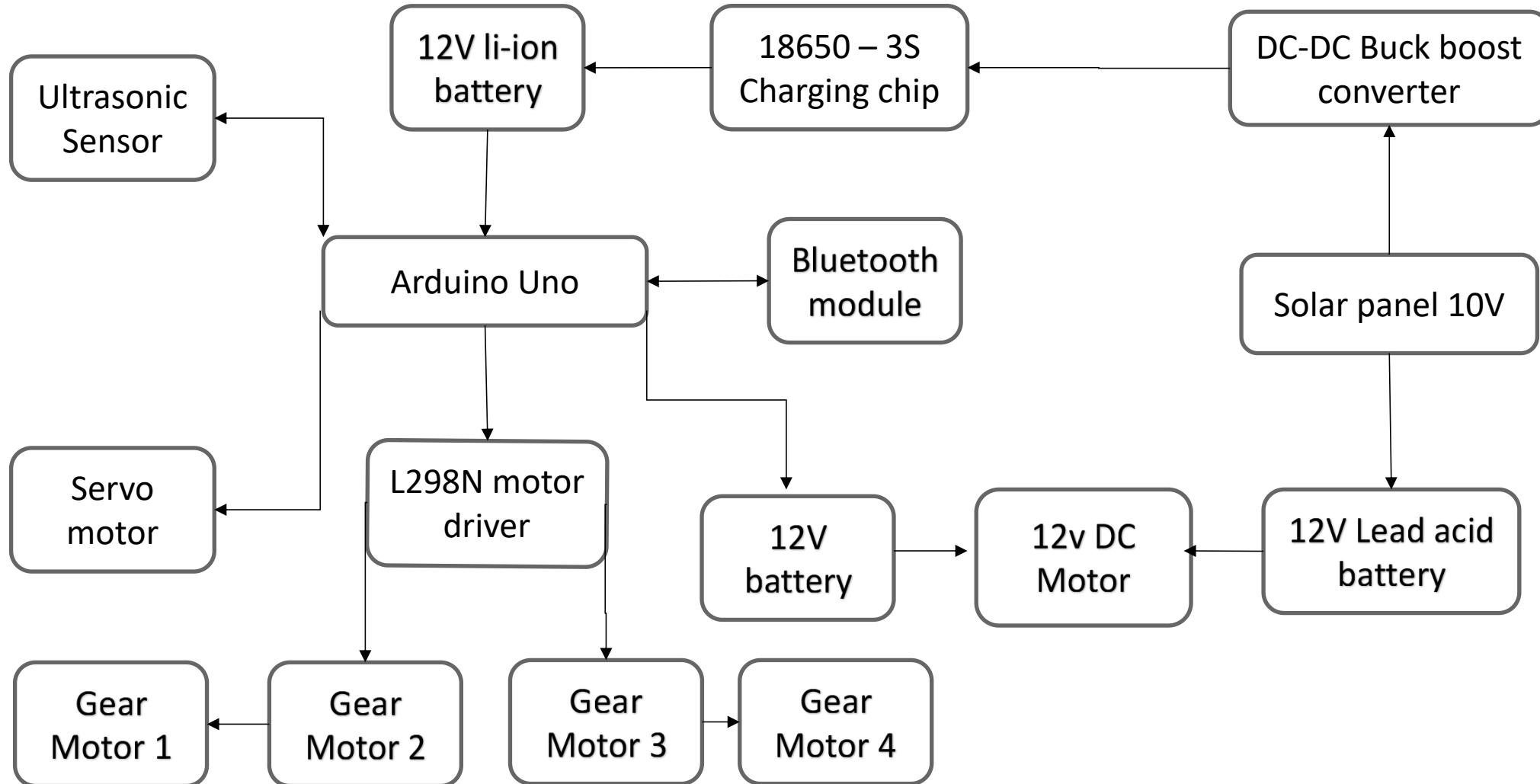
# PROPOSED SYSTEM-DESCRIPTION

- The proposed system consists of an IoT-enabled rechargeable electric lawn mower that can be controlled via a mobile app or web interface.
- The mower is powered by a battery-operated motor and uses sensors for obstacle detection, and real-time monitoring.

## **Key features include:**

- Automation through IoT Rechargeable battery for eco-friendliness and cost savings.
- Obstacle detection sensors for autonomous movement.
- Cutting height adjustment for better lawn maintenance.
- Solar charging integration for extended operation

# BLOCK DIAGRAM



# METHODOLOGY

1. **System Design:** The overall architecture is designed, including the mower body, wheel configuration, blade mechanism, and control system layout.
2. **Sensor Integration:** Ultrasonic sensors are integrated to detect obstacles, while GPS or boundary wires ensure the mower stays within the defined area.
3. **Motor & Blade Control:** The motor drivers control the wheel motors for movement and blade motors for cutting.
4. **Testing & Optimization:** The complete system is tested on various lawn types to ensure efficient grass cutting, reliable obstacle avoidance, and optimized battery performance.

# COST ANALYSIS

| component                  | Estimated cost |
|----------------------------|----------------|
| Arduino Uno                | 350-500        |
| DC-DC Buck Boost Converter | 300-600        |
| Ultrasonic Sensor          | 100-200        |
| Bluetooth Module           | 200-400        |
| Solar panel(24v)           | 2500-3500      |
| Lead Acid battery(12v)     | 1000-2000      |
| Li-ion battery(12v)        | 800-1500       |
| 18650 - 3s Charging Chip   | 100-200        |
| Bread board & wires        | 100-200        |
| Gear Motor(4 pcs)          | 1200-2000      |
| L298N Motor Driver         | 150-300        |
| Servo Motor                | 300-600        |
| Total cost (approximately) | 7100-12000     |



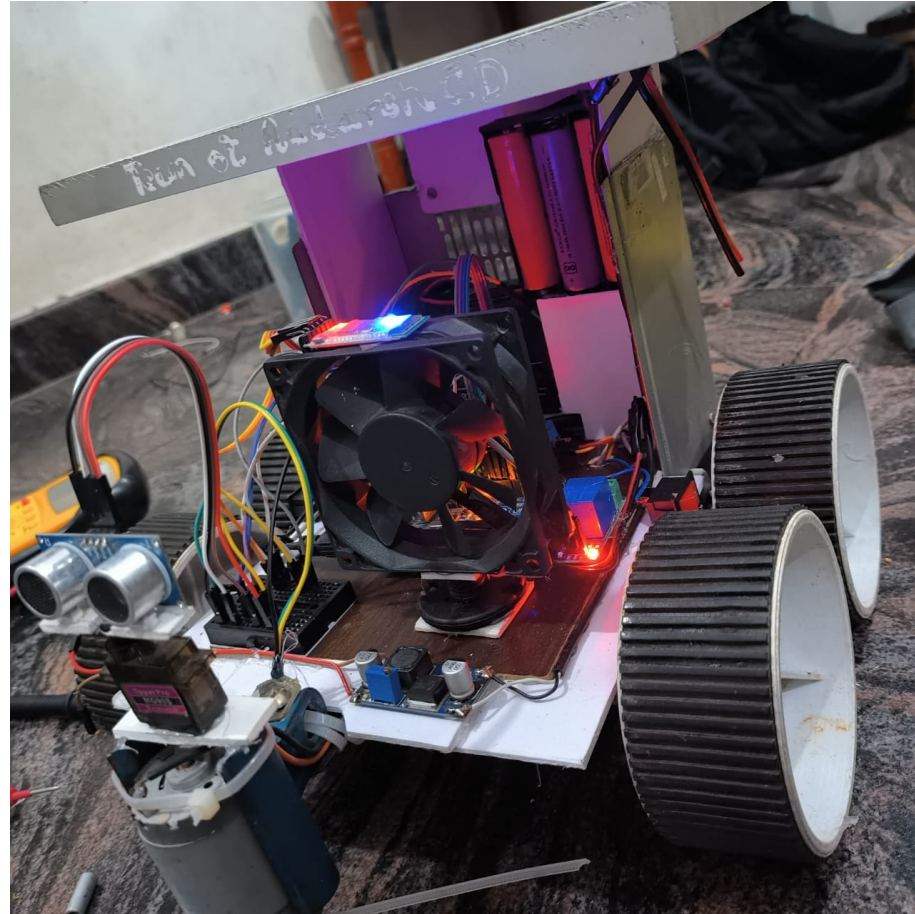
# FUTURE WORK

- 1.Voice Control and Smart Assistant Integration.
- 2.Automated Charging Dock.
- 3.Weather-Adaptive Mowing.
- 4.Multi-Terrain Adaptability.

# RESULT AND DISCUSSION

The autonomous lawn mower is an eco-friendly solution designed to automate the grass cutting process with minimal manual effort. Powered by a 12V battery supported by a 24V solar panel, the system uses an Arduino Uno to control multiple gear motors and a DC motor for efficient lawn mowing. Ultrasonic sensors help in obstacle detection, ensuring smooth navigation, while the Bluetooth module allows for easy manual override if needed. This smart lawn mower not only saves time and labor but also promotes green energy through its solar charging mechanism.

# TESTING MODEL



# REFERENCES

- [1].Jackson, L and R Hall. "Autonomous Lawn Mower Path Planning Using Genetic Algorithms". Computers & Operations Research 110 (2020): 147-156.
- [2].Martinez, C and R Lopez. "Collaborative Autonomous Lawn Mowing Using Swarm Robotics". IEEE Robotics and Automation Letters 8.1 (2023): 145-153.
- [3].Parekh, D, et al. "" A Review on Autonomous Vehicles: Progress, Methods and Challenges. Electronics 11 (2022):
- [4].Jackson, L and R Hall. "Autonomous Lawn Mower Path Planning Using Genetic Algorithms". Computers & Operations Research 110 (2020): 147-156.
- [5].Ibrahim, A and S Lee. "A Comparative Study of Autonomous Lawn Mower Navigation Techniques". Journal of Intelligent & Robotic Systems 76.3-4 (2019): 567-578.

**THANK YOU**